

Problem Set 1: Risk and Return

Investments

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Q1

You just bought 100 shares of ABC Inc. at \$100 per share. ABC paid out a dividend just before you bought the share and will pay a dividend of \$2 per share with certainty. The price of ABC a year from now, after ABC has paid out the dividend, is, however, uncertain and depends on the state of the economy. A year from now the economy will either be in a recession, a state of “normal” growth, or a boom with probabilities of 30%, 40%, and 30% respectively. After analyzing ABC you determine that the price of ABC a year from now in these various states of the economy will be:

State of the Economy	Price of ABC
Recession	\$80
Normal Growth	\$110
Boom	\$130

What is the expected return over the next year on your investment in ABC?

What is the standard deviation of the return?

The return in each state, with $r = \frac{P_{t+1} + D_{t+1} - P_t}{P_t}$:

$$r_{Recession} = \frac{80 + 2 - 100}{100} = -0.18$$

$$r_{Normal} = \frac{110 + 2 - 100}{100} = 0.12$$

$$r_{Boom} = \frac{130 + 2 - 100}{100} = 0.32$$

Given the return in each state, the expected return is:

$$E[r] = (-0.18)(0.30) + (0.12)(0.40) + (0.32)(0.30) = 0.09$$

To calculate the standard deviation, first calculate the variance:

$$\sigma^2[r] = 0.30(-0.18 - 0.09)^2 + 0.40(0.12 - 0.09)^2 + 0.30(0.32 - 0.09)^2 = 0.0381$$

Then the standard deviation is:

$$\sigma[r] = \sqrt{0.0381} \approx 0.195$$

Q2

Consider two stocks A and B. These stocks have returns that are driven by the state of the economy. Suppose that there are three states: “Recession,” “Normal,” “Boom.” These states occur with probabilities 20%, 50% and 30% respectively. The returns to the two stocks in the three states of the economy are given by:

State	Return to Stock A	Return to Stock B
Recession	-5%	2%
Normal	6%	5%
Boom	10%	12%

(a) What is the expected return of each stock?

The expected return is the probability-weighted sum of the returns in each state:

$$E[r_A] = (-0.05)(0.20) + (0.06)(0.50) + (0.10)(0.30) = 0.05$$

$$E[r_B] = (0.02)(0.20) + (0.05)(0.50) + (0.12)(0.30) = 0.065$$

(b) What is the standard deviation of each stock return?

To calculate the standard deviation, first calculate the variance of each stock:

$$\sigma^2[r_A] = 0.20(-0.05 - 0.05)^2 + 0.50(0.06 - 0.05)^2 + 0.30(0.10 - 0.05)^2 = 0.0028$$

$$\sigma^2[r_B] = 0.20(0.02 - 0.065)^2 + 0.50(0.05 - 0.065)^2 + 0.30(0.12 - 0.065)^2 = 0.001425$$

Then the standard deviation of each stock is:

$$\sigma[r_A] = \sqrt{0.0028} \approx 0.0529$$

$$\sigma[r_B] = \sqrt{0.001425} \approx 0.0377$$

Q3

Consider the stock ABC. ABC is currently paying a dividend of \$58. Given the riskiness of ABC, you believe a fair discount rate is 8%.

(a) What is the fair price of ABC if you expect dividends to grow at a constant rate of 5% in perpetuity?

With constant growth in perpetuity, using the Gordon growth model:

$$P_t = \frac{D_{t+1}}{r - g} = \frac{D_t \times (1 + g)}{r - g} = \frac{\$58 \times (1 + 0.05)}{0.08 - 0.05} = \$2,030$$

(b) What is the fair price of ABC if the growth rate is 5% for the next 5 years and then 3% afterwards?

The price is the present value of the first 5 dividends plus the present value of the price at the end of year 5:

$$P_t = \frac{D_{t+1}}{(1 + r)^1} + \frac{D_{t+2}}{(1 + r)^2} + \frac{D_{t+3}}{(1 + r)^3} + \frac{D_{t+4}}{(1 + r)^4} + \frac{D_{t+5}}{(1 + r)^5} + \frac{P_{t+5}}{(1 + r)^5}$$

First, calculating the dividends for the 5 years of 5% growth:

$$D_{t+1} = \$58 \times (1.05)^1 = \$60.90$$

$$D_{t+2} = \$58 \times (1.05)^2 = \$63.95$$

$$D_{t+3} = \$58 \times (1.05)^3 = \$67.14$$

$$D_{t+4} = \$58 \times (1.05)^4 = \$70.50$$

$$D_{t+5} = \$58 \times (1.05)^5 = \$74.02$$

After year 5, dividends grow at 3% in perpetuity. Thus the price at the end of year 5 comes from the Gordon growth model with $g = 3\%$:

$$P_{t+5} = \frac{D_{t+6}}{r - g} = \frac{D_{t+5} \times (1 + g)}{r - g} = \frac{\$74.0243 \times (1 + 0.03)}{0.08 - 0.03} = \$1,524.90$$

Then discounting each cash flow to today:

$$P_t = \frac{60.90}{1.08} + \frac{63.95}{1.08^2} + \frac{67.14}{1.08^3} + \frac{70.50}{1.08^4} + \frac{74.02}{1.08^5} + \frac{1,524.90}{1.08^5}$$

$$P_t = 56.39 + 54.82 + 53.30 + 51.82 + 50.38 + 1,037.82 \approx \$1,304.5$$

Q4

The stock market is trading at \$3,000 and paying a dividend of \$60. You expect the growth rates of dividends to be 4% in perpetuity.

(a) What is the expected return on the stock market?

Considering the constant expected returns and constant growth, the Gordon growth model applies:

$$P_t = \frac{D_{t+1}}{r - g} = \frac{D_t \times (1 + g)}{r - g}$$

Solving for the expected return r :

$$r = \frac{D_t \times (1 + g)}{P_t} + g$$

Then the expected return on the market is:

$$r = \frac{\$60 \times (1 + 0.04)}{\$3,000} + 0.04 = 0.0208 + 0.04 = 0.0608$$

The stock market suddenly drops to \$2,000. After careful analysis, you conclude that the expected growth rates of dividends is unchanged.

(b) What is the expected return on the market now?

$$r = \frac{\$60 \times (1 + 0.04)}{\$2,000} + 0.04 = 0.0312 + 0.04 = 0.0712$$

(c) What could potentially explain such a change in expected returns?

The expected dividends are the same before and after the drop. Thus the price decrease cannot come from lower expected cash flows. It must come from a higher discount rate: investors now demand a higher return (7.12%) to hold the market, and the same dividends discounted at a higher rate give a lower price.